(vii) $\frac{22}{7}$ is a _____number.

(viii) The conjugate of $-3+5i$ is _____.

(ix) In $2i(3-i)$, the real part is _____.

(x) The product of two complex numbers $(a,b)$ and $(c,d)$ i.e. $(a,b).(c.d)$ = _____.

**2. Read the following sentences carefully and encircle 'T' in case of true and 'F' in case of false statement.**

(i) $\mathbb{R}$ is closed under multiplication. T / F

(ii) if $x<y \wedge y<z \Rightarrow x<z$. T / F

(iii) $\forall x,y,z \in \mathbb{R}, x(y-z)=xy-xz$ T / F

(iv) The product of every two imaginary numbers is real. T / F

(v) The sum of two real numbers is a real number. T / F

**3. Tick (✓) the correct answer.**

(i) The additive inverse of $\sqrt{5}$ is

(a) $-\sqrt{5}$ (b) $\frac{1}{\sqrt{5}}$ (c) $\sqrt{-5}$ (d) $-5$

(ii) $(5i).(-2i)$ =

(a) $-10$ (b) $10$ (c) $-10i$ (d) $10i$

(iii) $3(5+7)=3.5+3.7$, name the property used

(a) Commutative (b) Associative (c) Distributive (d) Closure

(iv) $\sqrt{-2}\times\sqrt{-2}$ =

(a) $2$ (b) $-2$ (c) $2i$ (d) $-2i$

**4. Simplify** (i) $2^{3^2} \div 2^{2^3}$ (ii) $3^{2^3} \div 3^{3^2}$

**5. Simplify** (i) $3^{20}+3^{20}+3^{20}$ (ii) $2^{35}+2^{35}$

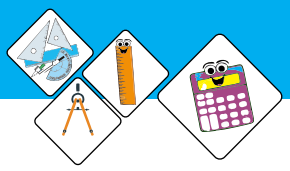
**6. Let z = 7-i, find**

(i) Re $(\bar{z})$ (ii) Im (z) (iii) $\bar{z}$ (iv) $|\bar{z}|$

(v) $z^{-1}$ (vi) $|z|^{-1}$ (vii) $iz$ (viii) $i\bar{z}$

## Summary

- The set of real numbers is the union of set of rational and irrational numbers i.e., $\mathbb{R}=Q\cup Q'$.
- There are two types of non–terminating decimal fractions i.e., non–recurring decimal fractions and recurring decimal fractions.



**3.** Tick (✓) the correct answers.

(i) If $\log_{10} x = 4$, then $x =$ _____ .

(a) 500 (b) 100 (c) 1000 (d) 10000

(ii) The characteristic of log 54.58 is_____ .

(a) 0 (b) 1 (c) 2 (d) 4

(iii) The base of common logarithm is_____ .

(a) 5 (b) 10 (c) e (d) 100

(iv) $\log xyz =$ _____ .

(a) $\log x \log y \log z$ (b) $\log x + \log y + \log z$

(c) $\log(xy)^z$ (d) $\log x - \log y - \log z$

(v) Scientific notation of 0.00789 is _____ .

(a) $7.89\times10^{-3}$ (b) $7.89\times10^{3}$

(c) $0.789\times10^{-2}$ (d) $78.9\times10^{-4}$

(vi) If $\log x = 2$ then $x =$_____ .

(a) 200 (b) 1000 (c) 100 (d) $\frac{2}{10}$

(vii) If $\log_2 8 = x$ then $x =$_____ .

(a) 64 (b) $3^2$ (c) 3 (d) $2^8$

(viii) Base in the Natural logarithm is _____ .

(a) 10 (b) $e$ (c) $\pi$ (d) 5

(ix) $3^5 = 243$, can be written in logarithmic form as_____ .

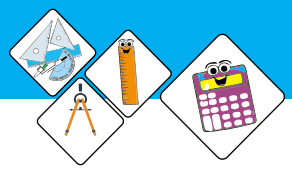
(a) $\log_3 5 = 243$ (b) $\log_3 243 = 5$

(c) $\log_5 243 = 12$ (d) $\log_5 3 = 243$

(x) If b>0 and b ≠ 1 then $\log_b \sqrt{b}$ =_____.

(a) 0 (b) 1

(c) $\frac{1}{2}$ (d) 2



## Review Exercise 3

**1.** **Encircle the correct answer.**

(i) Every polynomial is:

(a) an irrational expression (b) a rational expression
(c) a sentence (d) none of these

(ii) A surd which contains sum of two monomial surds is called

(a) Trinomial surd (b) Binomial surd
(c) Conjugate surd (d) Monomial surd

(iii) $3x + 2y - 3$ is an algebraic

(a) Expression (b) Equation
(c) Sentence (d) In-equation

(iv) The degree of the $3x^2y + 5y^4 - 10$ is

(a) 4 (b) 5 (c) 6 (d) 3

(v) $\sqrt{7}$ is an example of

(a) Monomial surd (b) Trinomial surd
(c) Binomial surd (d) Conjugate surd

(vi) Quotient $\frac{p(x)}{q(x)}$ of two polynomials $p(x)$ and $q(x)$, where $q(x) \neq 0$ is called

(a) Rational expression (b) Irrational expression
(c) Polynomial (d) Conjugate

(vii) $\frac{1}{x-y} - \frac{1}{x+y}$ is equal to

(a) $\frac{2x}{x^2-y^2}$ (b) $\frac{2y}{x^2-y^2}$ (c) $\frac{-2x}{x^2-y^2}$ (d) $\frac{-2y}{x^2-y^2}$

(viii) Conjugate of $2 - \sqrt{3}$ is

(a) $2+\sqrt{3}$ (b) $-2-\sqrt{3}$ (c) $\sqrt{2}+3$ (d) $\sqrt{3}-4$

(ix) $a^3 - 3ab(a-b) - b^3$ is equal to

(a) $(a-b)^3$ (b) $(a+b)^3$ (c) $a^3+b^3$ (d) $a^3-b^3$

(x) If $a+b=5$ and $a-b=3$, then the value of $ab$ is

(a) 4 (b) 5 (c) 3 (d) 6

(xi) $\left(5+\sqrt{15}\right)\left(5-\sqrt{15}\right)$ is equal to

(a) 10 (b) 15 (c) 25 (d) 30

(xii) $a^2+b^2+c^2+2ab+2bc+2ca$ is equal to

(a) $(a+b-c)^2$ (b) $(a+b+c)^2$

(c) $(a-b+c)^2$ (d) $(a+b+c)^3$

**2. Fill in the blanks.**

(i) Degree of any polynomial is _____ .

(ii) Conjugate of surd $2-\sqrt{3}$ is _____ .

(iii) Degree of polynomial $2x^3+x^2-4x^4+7x-9$ is _____ .

(iv) $\dfrac{\sqrt{x}}{3x+5}$ is a/an _____ expression.

(v) $(x-y)(x+y)\left(x^2+y^2\right)=$_____ .

## Summary

- A polynomial expression (simply say polynomial) in one variable $x$ can be written as: $a_0x^n+a_1x^{n-1}+a_2x^{n-2}+a_3x^{n-3}+\cdots+a_{n-1}x^1+a_n$
  A polynomial is usually denoted by $p(x)$.
- An algebraic expression which can be written in the form $\dfrac{p(x)}{q(x)}$, where $q(x) \neq 0$, and $p(x)$, $q(x)$ are both polynomials, called **rational expression** in $x$.
- An algebraic expression which cannot be written in form of $\dfrac{p(x)}{q(x)}$, where $q(x) \neq 0$, and $p(x)$, $q(x)$ are both polynomials, called irrational expression
- A polynomial expression consisting of only single term is called monomial.
- A polynomial expression consisting of two terms is called binomial.
- A polynomial expression consisting of three terms is called trinomial.
- Polynomial expression consisting two or more than two terms is called multinomial.

(iv) $a^4 - b^4 = (a-b)(a+b)(a+b)^2$ T/F

(v) $a^6 + b^6 = (a^3+b^3)(a^3-b^3)$ T/F

(vi) $a^5 + b^5 = (a+b)^5$ T/F

**2. Complete the following sentences**

(i) $16x^2 - y^4 = (4x - y^2)$ _____

(ii) $x^3 - 64y^3 = (x - 4y)$ _____

(iii) $x^2 + 5x + 6 = (x+2)$ _____

(iv) $x^2 + y^2 = (x - y)^2$ _____

(v) $a^3 + 27b^3 = (a + 3b)$ _____

**3. Tick (✓) the correct answers**

(i) Factors of $a^2 + 2a - 24$ are:

(a) $a+4, a-6$ (b) $a-4, a+6$

(c) $a+3, a-8$ (d) $a+8, a-3$

(ii) Factors of $a^2 + 2ab + b^2 - c^2$ is:

(a) $(a-b+c)(x-b-c)$ (b) $(a+b+c)(a-b-c)$

(c) $(a+b+c)(a+b-c)$ (d) $(a+b+c)(a-b-c)$

(iii) Factors of $x^3 + y^3$ is:

(a) $(x-y)(x^2+xy+y^2)$ (b) $(x+y)(x^2+xy+y^2)$

(c) $(x+y)(x^2+xy-y^2)$ (d) $(x+y)(x^2-xy+y^2)$

(iv) Factors of $y^3 - 27z^3$ are:

(a) $y-3z, y^2+3yz+9z^2$ (b) $y-3z, y+3z+9z^2$

(c) $y-3z, y^2-3yz+9z^2$ (d) $y+3z, y^2-3yz+9z^2$

(v) In simplified form $\frac{1}{x+y} + \frac{y}{x^2-y^2} =$

(a) $\frac{y+1}{x^2-y^2}$ (b) $\frac{x}{x^2-y^2}$

(c) $\frac{y}{x^2-y^2}$ (d) $\frac{y-1}{x^2-y^2}$

(vi) Find $m$, so that $x^2+4x+m$ is a complete square

(a) 16 (b) –16

(c) 4 (d) –4

## Review Exercise 5

**1. True and false questions**

Read the following sentences carefully and encircle 'T' in case of true and 'F' in case of false statement.

(i) HCF of $y^2 - 4$ and $y - 2$ is $y - 2$. T/F

(ii) HCF of $a^3 - 1$ and $a^2 - 1$ is $a + 1$. T/F

(iii) LCM of $x^3 + 1$ and $x + 1$ is $x^3 + 1$. T/F

(iv) LCM of $x^4 - y^4$ and $x^2 - y^2$ is $x^2 + y^2$. T/F

(v) HCF of $a^2 + 4a + 3$ and $a^2 + 5a + 6$ is $a + 3$. T/F

**2. Fill in the blanks.**

(i) There are _____ methods for finding the HCF of polynomials.

(ii) LCM × HCF of two polynomials $p(x)$ and $q(x)$ = _____.

(iii) HCF of $y^2 - 5y$ , $y + 6$ and $y - 2$ is _____.

(iv) LCM of $y^2+3y+2$ and $y^2+5y+6$ is _____.

(v) HCF of $y^2 - \frac{1}{y^2}$ and $y + \frac{1}{y}$ is _____.

**3. Tick (✓) the correct answers**

(i) HCF of $x^3 - 8y^3$ and $x^2 - 4xy + 4y^2$ is:

(a) $x - 4y$ (b) $x^2+2xy+y^2$

(c) $x+2y$ (d) $x-2y$

(ii) LCM of $(2y+3z)^5$ and $(2y+3z)^3$ is:

(a) $(2y+3z)^8$ (b) $(2y+3z)^3$

(c) $(2y+3z)^2$ (d) $(2y+3z)^5$

(iii) HCF of $x^3 - y^3$ and $x^2+xy+y^2$ is:

(a) $x+y$ (b) $x^2+xy+y^2$

(c) $x-y$ (d) $(x-y)^2$

(iv) LCM of $(x-y)^4$ and $(x-y)^3$ is:

(a) $(x-y)$ (b) $(x-y)^3$

(c) $(x-y)^4$ (d) $(x-y)^7$

(v) Simplified form of $\frac{1}{x+y}+\frac{y}{x^2-y^2}$ is:

(a) $\frac{y+1}{x^2-y^2}$ (b) $\frac{x}{x^2-y^2}$

(c) $\frac{y}{x^2-y^2}$ (d) $\frac{x+y}{x^2-y^2}$

(vi) Simplified form of $\frac{y}{25x^2-y^2}-\frac{1}{5x-y}$ is:

(a) $\frac{5x}{25x^2-y^2}$ (b) $\frac{5x}{5x-y}$

(c) $\frac{-5x}{5x+y}$ (d) $\frac{-5x}{25x-y^2}$

(vii) $\frac{a^3x^3+a^3y^3}{a^2(x+y)}$ =_____:

(a) $ax^2+ay^2$ (b) $x^2+y^2$

(c) $a(x^2-xy+y^2)$ (d) $a(x^2+xy+y^2)$

(viii) $\frac{a}{a-b}-\frac{b}{a+b}$ =_____:

(a) $\frac{a^2+b^2}{a-b}$ (b) $\frac{a^2+b^2}{a^2-b^2}$

(c) $\frac{a+b}{a^2-b^2}$ (d) $\frac{a-b}{a^2-b^2}$

(ix) LCM =_____, given that $p$ and $q$ are any two polynomials.

(a) $\frac{\text{HCF}}{p\times q}$ (b) $\frac{p\times q}{\text{HCF}}$

(c) $\frac{p}{\text{HCF}}$ (d) $\frac{q}{\text{HCF}}$

(x) LCM of $x^2-x+1$ and $x^3+1$ is:

(a) $x+1$ (b) $x^2-x+1$ (c) $x^3+1$ (d) $x^2+x+1$

## Review Exercise 6

**1. True and false questions**

Read the following sentences carefully and encircle 'T' in case of true and 'F' in case of false statement.

(i) $ay+b=0$, where $a=0$ is a linear equation T/F

(ii) The solution set of $3y-2<7, y\in\mathbb{N}$ is $\{4, 5, 6,\ldots\}$ T/F

(iii) The solution set of $\sqrt{y}+1=3$ is $\{4\}$. T/F

(iv) The solution set of $|4y|=8$ is $\{2,-2\}$. T/F

(v) The solution set of $-2\le x\le 2, x\in Z$ is $\{-2,0,2\}$. T/F

**2. Fill in the blanks.**

(i) The solution set of $2y=-y$ is_____.

(ii) The solution set of $\sqrt{y+5}=5$ is_____.

(iii) The solution set of $|x|-4=0$ is _____.

(iv) The solution set of $\sqrt{x+5}+2=4$ is_____.

(v) The solution set of $0<y+2<5$ when $y\in\mathbb{R}$ is_____.

**3. Tick (✓) the correct answer**

(i) The solution set of linear equation in one variable has

(a) One solution (b) Two solutions

(c) Three solutions (d) More than one solutions

(ii) $|-20|$

(a) $=20$ (b) $<20$

(c) $=-20$ (d) $>20$

(iii) $x\le 4$ means

(a) $x<4$ (b) $x=4$

(c) $x<4$ or $x=4$ (d) $x>4$ or $x=4$

(iv) The solution set of $\sqrt{y}=10$ is

(a) {100} (b) {10}

(c) {–10} (d) {–10,10}

(v) $\sqrt{y+4}+2=8$ is a

(a) Linear equation (b) Radical equation

(c) Cubic equation (d) Quadratic equation

(vi) The solution set of $5-3y=-7$ is

(a) $\{-4\}$ (b) $\{1, 4\}$

(c) $\{4\}$ (d) $\{12\}$

(vii) The solution set of $\sqrt{5y+5}+5=10$ is

(a) $\{\pm 4\}$ (b) $\{5\}$

(c) $\{4\}$ (d) $\{-4\}$

(viii) The solution set of $\left|\frac{5y}{3}\right|=5$ is

(a) $\{3\}$ (b) $\{-5, 5\}$

(c) $\{3, -3\}$ (d) $\{-3\}$

(ix) The solution set of $|-y|=0$ is

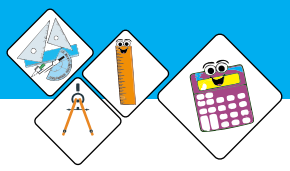
(a) $\{1\}$ (b) $\{-1\}$

(c) $\{0\}$ (d) $\{\ \}$

(x) If $x>0$, $y>0$ and $x-y<0$, then which of the following relation holds good?

(a) $x<y$ (b) $x+y<0$

(c) $x>y$ (d) $y-x<0$



## Review Exercise 7

**True and false question**

**1.** Read the following sentences carefully and encircle T or F whichever is.

(i) The Cartesian plane is also called $xy$-plane. T / F

(ii) In 2nd quadrant both $x$ and $y$ coordinates are positive. T / F

(iii) The point (1, –2) lies in 1st quadrant. T / F

(iv) The (–3, –4) lies in the 4th quadrant. T / F

**2.** Tick (✓) the correct answer in the following:

(i) The point (–3, –4) is located in

(a) 1st quadrant (b) 2nd quadrant

(c) 3rd quadrant (d) 4th quadrant

(ii) The two coordinates axes intersect at an angle of

(a) $45^0$ (b) $90^0$

(c) $180^0$ (d) $270^0$

(iii) The line y = 4 is parallel to

(a) $x$-axis (b) $y$-axis

(c) Both axes (d) None

(iv) The line $x$=–5 is parallel to

(a) $x$-axis (b) $y$-axis

(c) Both axis (d) None

(v) The line $x$=–5 has a point on $x$-axis

(a) (–5, 5) (b) (0, -5)

(c) (–5, 0) (d) (5, 0)

(vi) The solution set of the line $x$=2 and $x$=5

(a) {(2, 5)} (b) {2, 5}

(c) {(0, 5)} (d) { }

(vii) The co-ordinate axes are mutually

(a) Perpendicular (b) Parallel

(c) Intersecting at $30^o$ (d) Intersecting at $45^o$

## Review Exercise 8

**1. Fill in the blanks**

(i) A polynomial equation in which degree of variable is _____ called quadratic equation.

(ii) Standard form of quadratic equation is _____ .

(iii) $3^x + 3^{2x} = 1$ is called _____ equation.

(iv) Solution of $3^x = 9$ is_____.

(v) Solution of $ax^2 + bx + c = 0$ is _____ .

**2. Tick (✓) the correct answer**

(i) Degree of quadratic equation is

(a) 1 (b) 2 (c) 3 (d) 4

(ii) Standard form of quadratic equation is

(a) $ax^2 + bx + c = 0, a \neq 0$ (b) $ax^2 + c = 0, a \neq 0$

(c) $ax^2 + bx = 0, a \neq 0$ (d) $ax^3 + bx^2 + c = 0, a \neq 0$

(iii) The Quadratic Formula for $ax^2 + bx + c = 0, a \neq 0$ is

(a) $x = \frac{-b - \sqrt{b^2 - 4ac}}{2}$ (b) $x = \frac{b \pm \sqrt{b^2 - 4ac}}{2a}$

(c) $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ (d) $x = \frac{-b \pm \sqrt{b^2 + 4ac}}{2a}$

(iv) Solution set of $x^2 + 10x + 24 = 0$ is

(a) {–6 , –4 } (b) {–6 , 4 }

(c) { 6 , 4} (d) {6 , –4 }

(v) How many maximum roots of quadratic equation are

(a) 2 (b) 3 (c) 1 (d) 4

(vi) Two linear factors of $x^2 - 15x + 56$ are

(a) $(x-7)$ and $(x+8)$ (b) $(x+7)$ and $(x-8)$

(c) $(x-7)$ and $(x-8)$ (d) $(x+7)$ and $(x+8)$

(vii) Polynomial equation, which remains unchanged when $x$ is replaced by $\frac{1}{x}$ is called a/ an

(a) Exponential equation (b) Reciprocal equation

(c) Radical equation (d) none of these

(viii) An equation of the type of $3^x + 3^{2-x} + 6 = 0$ is a/an

(a) Exponential equation (b) Radical equation
(c) Reciprocal equation (d) none of these

(ix) The solution set of equation $4x^2 - 16 = 0$ is
(a) $\{\pm 4\}$ (b) $\{4\}$ (c) $\{\pm 2\}$ (d) none of these

(x) An equation of the form $2x^4 - 3x^3 + 7x^2 - 3x + 2 = 0$ is called a/an

(a) Reciprocal equation (b) Radical equation
(c) Exponential equation (d) none of these

**3. True and false questions**

Read the following sentences carefully and en-circle 'T' in case of true and 'F' in case of false statement.

(i) Every quadratic equation can be solved by factorization. T/F

(ii) Every Quartic equation has two roots. T/F

(iii) Every Quadratic equation can have no solution. T/F

(iv) $ax^2 + bx + c = 0$ is called the quadratic equation in $x$ if $a$=0 and $b$, $c$ are real numbers. T/F

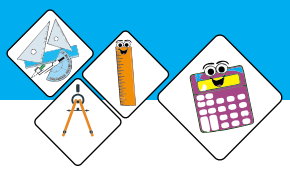
(v) Extraneous root satisfy the equation. T/F

(vi) Extraneous roots do not satisfy the equation. T/F

(vii) In the quadratic equation the highest exponent of the variable is two. T/F

## Summary

- A Polynomial equation in which degree of a variable is 2, called quadratic equation.
- $ax^2 + bx + c = 0$, $a \neq 0$, $a$, $b$, $c$ are real numbers is called standard form of a quadratic equation.
- Formula for quadratic equation $ax^2 + bx + c = 0$, $a \neq 0$ is $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- In exponential equations, variables occur in exponents.
- An equation in which the variable appears under the radical sign is called a radical equation.



**4.** Encircle the corresponding letters a,b,c or d for correct answer:

(i) Which of the following is not a sufficent condition for congurence of two triangles?

(a) A.S.A ≅A.S.A (b) H.S ≅ H.S

(c) S.A.A≅ S. A. A (d) A.A.A≅ A. A. A

(ii) In ΔABC, if ∠A ≅ ∠B, then the bisector of _____ angle divides the triangle into congruent triangles:

(a) ∠A (b) ∠B

(c) ∠C (d) any one of its angles.

(iii) The diagonal of ____ does not divide it into two congruent triangles:

(a) Rectangle (b) Trapezium

(c) Parallogram (d) Square

(iv) How many acute angles are there in an acute angled triangle?

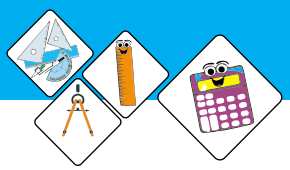
(a) 1 (b) 2

(c) 3 (d) not more than 2.

## Summary

In this unit we stated and proved the following theorem:

- In any correspondence of two traingles, if one side and any two angles of one traingle are congruent to the corresponding side and angles of the other, the two taringles are congruent. (A.S.A. ≅ A.S.A)
- If two angles of a traingles are congruent, then the side opposite to them are also congruent.
- In the correspondence of two traingles, if three sides of two traingles are congruent to the corresponding three sides of other, then the two traingles are congruent (S.S.S ≅ S.S.S).
- If in the correspondence of the two right-angled traingles the hypotenuse and one side of one traingles are congruent to the hypotenuse and the corresponding side of other, then the traingles are congruent. (H.S ≅ H.S).
- Two traingles are said to be congruent, if there exists a correspondence between them such that all the corresponding sides and traingles are congruent.(S.S.S).



**2. Tick (✓) the correct answer.**

(i) Diagonals of a square are _____ to each other.
- a) Perpendicular
- b) Non congruent
- c) Congruent
- d) Both 'a' and 'c'

(ii) Sum of the measures of interior angles of a quadrilateral is
- a) 2 right angles
- b) 4 right angles
- c) 3 right angles
- d) none of these

(iii) Measure of a line segment joining the mid points of $\overline{AB}$ and $\overline{AC}$ of $\Delta ABC$ is 3.5cm, then $m\overline{BC}$ =
- a) 4.5cm
- b) 5.5cm
- c) 6cm
- d) 7cm

(iv) Two medians $\overline{AD}$ and $\overline{BE}$ of $\Delta ABC$ intersect each other at G. If $m\overline{GD}$ = 1.7cm, then $m\overline{AG}$ =
- a) 2.7cm
- b) 8.85cm
- c) 3.4cm
- d) 5.1cm

(v) If sum of the measures of $\angle A$ and $\angle C$ of a parallelogram ABCD is 130°, then $m\angle B$=
a) 25° b) 115° c) 65° d) none of these

(vi) If opposite angles of a quadrilateral are equal in measures and none of them is a right angle, then the quadrilateral is a
- a) Square
- b) Parallelogram
- c) Trapezium
- d) Rectangle

(vii) Centroid is the common point of intersection of
- a) Medians of a triangle
- b) Diagonals of a parallelogram
- c) Angle bisectors of a triangle
- d) Perpendicular bisectors of a triangle

(viii) A point on median of a triangle is
- a) equidistant from its vertices
- b) equidistant from the mid points of its sides
- c) equidistant from its altitudes
- d) none of these

## Exercise 11.3

1. Two isosceles triangles have a common base, prove that the line joining vertices bisects the common base at right angle.
2. If the bisector of an angle of a triangle bisects the opposite side, prove that triangle is an isosceles.
3. In an isosceles $\Delta ABC$, $m\overline{AB} = m\overline{AC}$. Prove that the perpendiculars from the vertices B and C to their opposite sides are equal.

## Review Exercise 11

1. Prove that, if two altitudes of a triangle are congruent, the triangle is an isosceles.
2. Prove that, a point in the interior of a triangle is an equidistant from all the three sides' lies on the bisector of all the three angles of the triangle.
3. Write **'T'** for True and **'F'** for False in front of each of the following statements
   (i) Bisection of side means, we divide the given side into two equal parts.
   (ii) In a right angled isosceles triangle each angle on the base is of 45° .
   (iii) Triangle of congruent sides has congruent angles.
4. Choose the correct option:

(i) There are _____ acute angles in an acute angled triangle.
(a) One (b) Two (c) Three (d) None

(ii) An point equidistant from the end points of a line segment is on the _____ of it.
(a) Right bisector (b) Perpendicular
(c) Centre (d) Mid-point

(iii) _____ of the sides of an acute angled triangle intersect each other inside the triangle
(a) Perpendicular (b) The right bisector
(c) Obtuse (d) Acute

(v) The bisector of the angles of a triangle are _____.
(a) Concurrent (b) Collinear
(c) Do not interest (d) Unequal

**3. Tick (✓) the correct answer.**

(i) Measure of one side of an equilateral triangle is 6 cm, then the length of its median is _____ 9cm.

a) less than  b) greater than

c) equal to  d) none of the above

(ii) Perimeter of a rectangle is 22cm, then the length of its diagonal is _____ 11cm.

a) equal to  b) greater than

c) less than  d) none of the above

## Summary

- If two sides of a triangle are unequal in length, the longer side has an angle of greater measure opposite to it.
- If two angles of a triangle are unequal in measure, the side opposite to the greater angle is longer than the side opposite to the smaller angle.
- The sum of the lengths of any two sides of a triangle is greater than the length of the third side.
- From a point, outside a line, the perpendicular is the shortest distance from the point to the line.

iv) The perpendicular bisectors of the sides of a triangle are _____

v) Two or more than two triangles are said to be congruent if they are equiangular and measures of their corresponding sides are _____

**2. Tick (✓) the correct answer.**

**i)** A triangle having all the three sides congruent is called _____ triangle.

(a) scalene (b) right angled
(c) equilateral (d) isosceles

**ii)** A quadrilateral having each angle equal to 90° and all the sides are congruent is called _____

(a) parallelogram (b) rectangle
(c) trapezium (d) square

**iii)** The medians of a triangle are _____

(a) collinear (b) congruent
(c) concurrent (d) parallel

**iv)** The _____ altitudes of an equilateral triangle are congruent.

(a) two (b) three
(c) four (d) none

**v)** The diagonals of a rectangle _____ each other.

(a) bisect (b) trisect
(c) bisect at right angle (d) none of these

**vi)** The _____ of a triangle cut each other in the ratio 2:1.

(a) Altitudes (b) Angle bisectors
(c) Right bisectors (d) Medians

**vii)** If each angle on the base of an isosceles triangle is 45°, then the measure of the third angle is _____

(a) 30° (b) 60°
(c) 90° (d) 120°

**viii)** If the three medians of a triangle are congruent then the triangle is _____

(a) Right angled (b) equilateral
(c) Isosceles (d) acute angled

**ix)** If two _____of a triangle are congruent, then the triangle will be isosceles.

(a) Altitudes (b) Medians
(c) Right bisectors (d) sides

## Review Exercise 14

**1. Mark 'T' for True and 'F' for False in front of each given below:**

(i) Area of a closed figure means region enclosed by bounding lines of the figure. **T/F**

(ii) A diagonal of rectangle divides it into two congruent triangles. **T/F**

(iii) Congruent figures have different areas. **T/F**

(iv) The area of parallelogram is equal to the product of base and height. **T/F**

(v) Median of a triangle means perpendicular from a vertex to the opposite side (base). **T/F**

(vi) Perpendicular distance between two parallel lines can sometimes be different. **T/F**

(vii) An altitude drawn from a vertex always bisects the opposite side. **T/F**

(viii) Two triangles are equal in areas, if they have the same base and equal altitude. **T/F**

**2. Tick (✓) the correct answer.**

(i) If perpendicular distance between two lines is the same. The lines are _____

(a) Perpendicular to each other (b) Parallel to each other
(c) Intersecting to each other (d) None of these.

(ii) If two triangles have equal areas then they will _____be congruent as well.

(a) Not necessarily (b) Necessarily
(c) Definitely (d) None of these.

(iii) Perpendicular from a vertex of a triangle to its opposite side is called _____.

(a) Median (b) Perpendicular bisector
(c) Altitude (d) Angle bisector

(iv) Parallelograms having same base and same altitude are _____.

(a) Congruent (b) Equal in areas

(c) Similar (d) All of these.

(v) Two parallelograms have equal bases. They will be having the same area, if _____.

(a) Their altitudes are equal

(b) Their altitude is the same

(c) They lies between the same parallel lines

(d) All of these.

(vi) $\Delta ABC$ and $\Delta DEF$ have equal bases and equal altitudes, then triangles are_____.

(a) Equal in area (b) Congruent

(c) Similar (d) None of these.

## Summary

- In this unit we have mentioned some necessary preliminaries, stated and proved the following theorems along with corollaries, if any.
- Parallelograms on the same base and between the same parallel lines (or of the same altitude) are equal in areas.
- Parallelograms on the equal bases and having the same (or equal) altitude are equal in areas.
- Triangles on the same base and of the same (i.e. equal) altitudes are equal in areas.
- Triangles on equal bases and of equal altitudes are equal in areas.
- Area of a figure means region enclosed by the boundary lines of a closed figure.
- A triangular region means the union of triangle and its interior.
- By area of triangle means the area of its triangular region.
- Altitude or height of a triangle means perpendicular distance to base from its opposite vertex.

## Review Exercise 16

**1.** Read the following sentences carefully and encircle "T" in case of True and "F" in case of False statement.

(i) $R$ is the mid-poin of $\overline{PQ}$, if $R$ is lying between $P$ and $Q$ T/F
(ii) In scalene triangle all sides are equal T/F
(iii) Perpendicular lines meet at an angle of 135°. T/F
(iv) Collinear points may form a triangle. T/F
(v) Non-collinear points form a triangle. T/F
(vi) In an isosceles triangle, two sides angles are equal. T/F
(vii) All the points that lie on the y-axis are collinear T/F
(viii) Intersection of $x$-axis and $y$-axis is (0,0) T/F
(ix) The distance from origin to (6,0) is 36 unit. T/F

**2.** Fill in the blanks.

(i) If $A(x_1, y_1)$ and $B(x_2, y_2)$ be the any two points on the line, then $|\overline{AB}|$= _____
(ii) Collinear points lies on the same _____
(iii) In 4th quadrant $x>0$ and $y$_____

**3.** Tick (✓) the correct answer.

(i) Two perpendicular lines meet at an angle of:
(a) 45° (b) 60°
(c) 90° (d) 180°

(ii) $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$ is called:
(a) Mid-point formula (b) Distance formula
(c) Division formula (d) Ratio formula

(iii) $A(3, 0)$ and $B(0, 3)$ are any two points in the plane then $|\overline{AB}|$=
(a) 6 unit (b) $6\sqrt{2}$ unit
(c) $3\sqrt{2}$ unit (d) $3\sqrt{2}$ unit

(iv) Three points A,B and C are collinear if
(a) $m\ \overline{AB} = m\ \overline{BC} + m\ \overline{AC}$ (b) $(m\ AB)^2 = (m\ BC)^2 + (m\ AC)^2$
(c) both a and b (d) $(m\ \overline{AB}) \neq m\ \overline{BC} + m\ \overline{AC}$